Exploring the interplay between Planetary Boundaries and Sustainable Development Goals using Large Language Models

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By analyzing 40,037 climate articles using Large Language Models (LLMs), we identified interactions between Planetary Boundaries (PBs) and Sustainable Development Goals (SDGs). An automated reasoner distinguished true trade-offs (SDG progress harming PBs) and synergies (mutual reinforcement) from double positives and negatives (shared drivers). Results show 21.1% true trade-offs, 28.3% synergies, and 19.5% neutral interactions, with the remainder being double positive or negative. Key findings include conflicts between land-use goals (SDG2/SDG6) and land system boundaries (PB6), together with the underrepresentation of social SDGs in the climate literature. Our study highlights the need for integrated policies that align development goals with planetary limits to reduce systemic conflicts. We propose three steps: (1) integrated socio-ecological metrics, (2) governance ensuring that SDG progress respects Earth system limits, and (3) equity measures protecting marginalized groups from boundary compliance costs.

Planetary Boundaries | Artificial Intelligence | Sustainable Development Goals | Large Language Models

The Planetary Boundary (PBs) framework (1) defines the limits of human activities required to maintain the stability of a stable and resilient Earth (2). They establish a "safe operating space" for humanity, beyond which the risk of generating abrupt or irreversible large-scale environmental changes increases. The nine identified boundaries encompass key Earth system processes, such as climate change and biogeochemical flows, emphasizing the interconnections between these critical processes and the need for a holistic approach to sustainability.

Since its introduction, the PB framework has been widely used to assess the impact of various research fields on the Earth system. For instance, the Rockefeller Foundation–Lancet Commission on Planetary Health (3) employs the PB concept to explore the intricate relationship between human health and the natural systems of the planet. Their research identifies several critical environmental changes that pose risks to human well-being, including climate change, ocean acidification, land degradation, water scarcity, overfishing, and loss of biodiversity.

On the other hand, the Sustainable Development Goals (SDGs), established by the United Nations in 2015, consist of 17 global objectives designed to eradicate poverty, protect the planet, and foster prosperity for all (4). These goals were set to be achieved by 2030, but it has become increasingly evident that many of the SDGs will not be met.

In a recent study, Vinuesa et al. (5) examined the role of artificial intelligence in advancing SDGs, highlighting the need for regulatory guidance. Their work inspired the present study, which builds on the idea that developing an integrated sustainability agenda, one that effectively addresses both social and environmental priorities, requires a deep understanding of the complex interplay between PBs and the SDGs.

Methodology

Data were extracted from 40,037 open-access climate articles using five sequential LLM queries per document (6, 7). The first four queries identified relevant SDGs and PBs, classified their interactions into three categories (synergy, trade-off, or neutral), and assessed the direction of influence between each pair. The fifth query performed a validation step, refining the initial classifications. It distinguished between true synergies (TS), where SDGs and PBs positively interact, and mere

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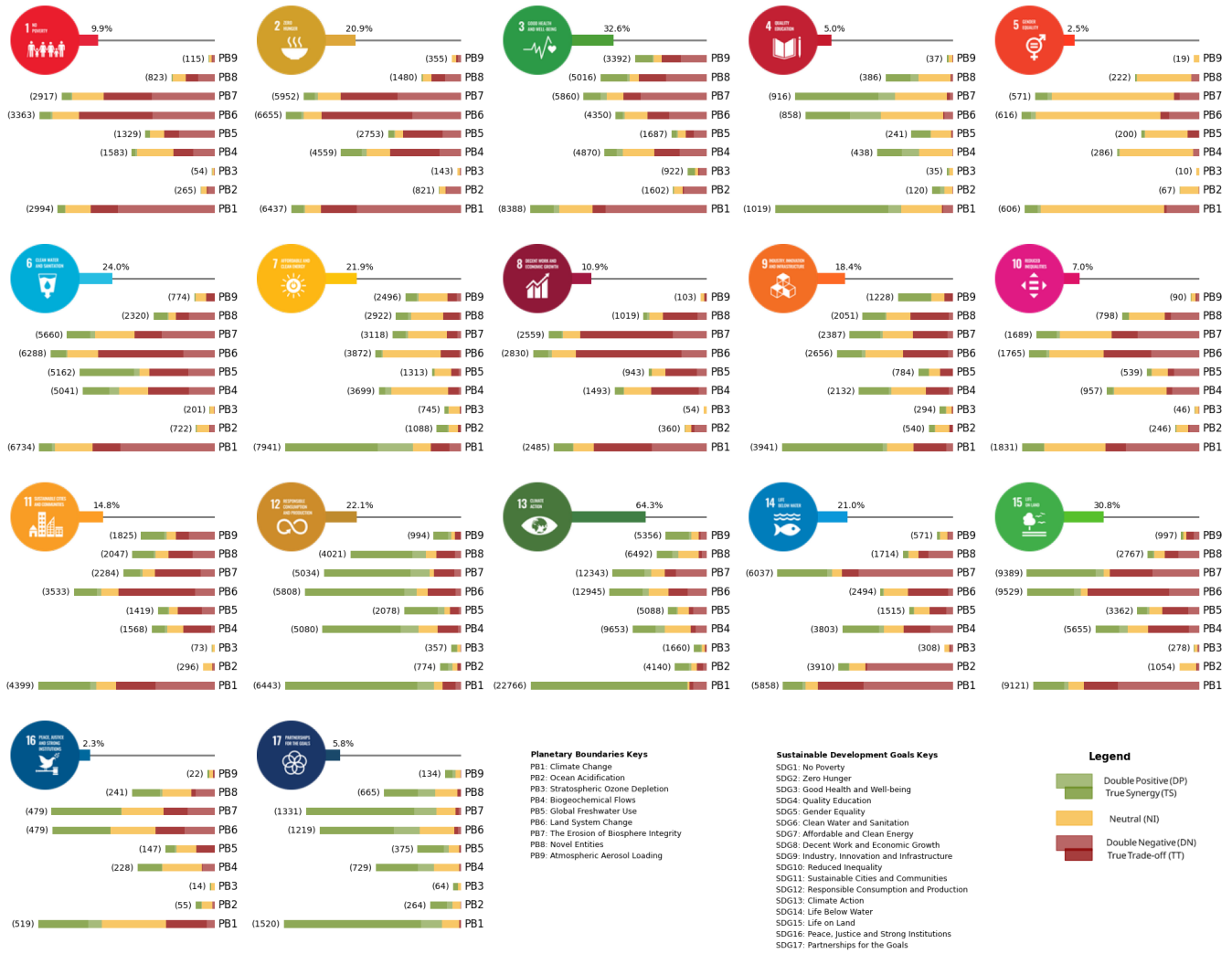

Fig. 1. Connections between the SDGs and PBs. Each subpanel shows interactions between one of the seventeen SDGs and the nine PBs. For each SDG, the top bar indicates the proportion of papers in the database related to that SDG. The stacked bars below represent the proportions of synergies (green), neutral interactions (beige), and trade-offs (red) for each PB. Darker overlays within these bars denote proportions of validated 'true' synergies (TS) or trade-offs (TT), distinguishing them from double positives or double negatives (lighter overlays). Bars are normalized by the maximum number of interactions per SDG, and the total number of interactions is indicated to the left of each bar.

positivity (DP), where the interaction may appear positive but lacks textual evidence within the document. Similarly, it differentiated true trade-offs, TT, (where improving one SDG increases pressure on a PB) from double negatives (DN), where simultaneous negative impacts occur on both the SDG and PB, but without an inverse relationship. DN were classified as trade-offs initially by the LLM, but the fifth query enabled a better classification. Further details on the methodology are provided in the SI.

Results. Figure 1 summarizes the quantitative SDG-PB interactions. The main panel features 17 plots, one per SDG, each with nine bars representing PBs. Labels show link counts, and stacked areas within bars depict the proportions of synergies (green), neutral (yellow), and trade-offs (red). Initial analysis showed 33.8% synergies, 44.9% trade-offs, and 19.5% neutral interactions (NI) overall. Our Reasoner refined this (darker overlays), identifying 28.3% as validated TS (mutual reinforcement) and 21.1% as TT (SDG progress increases the PB pressure).

Figure 1 highlights the disproportionate trade-offs between PB2 and SDG14, with 75.5% of PB2-SDG14 links classified as trade-offs—1.6 times the global average. Given PB2's low presence (12%) and SDG14's moderate coverage (21%), this contrast is interesting. Since improving SDG14 should not worsen PB2, and vice versa, the reasoner identified 97.5% of these cases as DN, where ocean acidification and marine ecosystem decline are driven by the same external pressures. This highlights a critical policy insight: interventions must address the root causes, particularly anthropogenic CO2 emissions, which degrade marine ecosystems (SDG14) and breach ocean acidity boundaries (PB2). Focusing on SDG14 or PB2 in isolation, without tackling shared drivers such as climate change, will likely prove insufficient for achieving sustainability in the marine environment.

Second, Land System Change (PB6), present in 42.1% of the documents, showed frequent conflict, with 52.5% of its

SDG interactions involving trade-offs (including DN), versus 25.3% synergies and 19.4% NI. Notably, SDG12 (Responsible consumption and Production) deviated strongly, exhibiting 69.2% synergy (88.8% TS), indicating that sustainable production may mitigate land degradation. For instance, regulated tourism and eco-certified infrastructure in Kazakhstan have reduced soil erosion and habitat fragmentation (8), a finding correctly identified by our LLM.

Our data revealed frequent conflicts between ensuring food security (SDG2 – Zero Hunger), clean water (SDG6 – Clean Water and Sanitation, and preserving land systems (PB6), with trade-offs identified in 78.1% and 70.0% of cases, respectively. This reflects the pressure on land quality from agricultural expansion and water demand, complicating the balance between resource use and sustainability. Examples include maize expansion impacting land cover (9) and EU biomass imports driving deforestation (10).

Despite thematic links, SDG15 (Life on Land) registered 62.4% trade-offs with PB6 (73.3% TT), indicating conflicts between habitat protection and extractive land use, such as altered ecosystems from adaptive forest management (11). SDG13 (Climate Action) showed context-dependent effects on PB6, with nearly equal synergies (38.5%, 84.3% TS) and trade-offs (38.0%, comprising 50.2% TT and 24.1% DN). The LLM flags consistent trade-offs from biofuel policies, which can displace crops and ecosystems (12).

Third, environment and economy-related SDGs dominate interactions, while social goals such as SDG16 (Peace, Justice and Strong Institutions, 2.3%), SDG5 (Gender Equality, 2.5%), and SDG4 (Quality Education, 4.0%) are underrepresented in climate literature, yet show 40% trade-offs when linked. Economic activities often framed as poverty reduction (e.g., agricultural intensification, resource extraction, and urbanization) can exacerbate PB pressures and disproportionately harm marginalized groups. Examples include resource extraction depleting reserves and displacing communities under weak governance (13), and urbanization driving biosphere degradation alongside social inequalities (14).

Directionality analysis revealed that 69.4% of the interactions were driven by PB-to-SDG pressures. However, industrial/energy SDGs (7, 9, 12) predominantly drive PB impacts, likely due to their focus on proactive interventions versus other SDGs responding to environmental pressures. For instance, SDG7's biofuel expansion can increase land pressure, SDG9's infrastructure may alter land/water systems, and SDG12's initiatives could unintentionally overuse resources. This suggests a potential divide between impact-driving and mitigating SDGs, although interactions are highly context-dependent and require full dataset analysis for comprehensive understanding.

Discussion. We analyzed SDG-PB synergies and trade-offs using Gemini 1.5 Flash, with a reasoner differentiating true trade-offs (improving one worsens another) from double negatives (both decline together) and validating synergies. This yielded 21.1% true trade-offs, 28.3% synergies, 19.5% neutral interactions, with the remainder being double synergies and trade-offs.

Ocean Acidification (PB2) and Life Below Water (SDG14) are frequently presented as double negatives, declining together due to shared external pressures rather than reinforcing each other synergistically. This underscores a critical policy need: tackling the root causes, particularly anthropogenic CO_2 emissions, is essential for simultaneously improving marine ecosystem health and respecting ocean acidity limits. Land System Change (PB6) conflicts with Zero Hunger (SDG2) and Clean Water and Sanitation (SDG6), showing 78.1% and 70.0% trade-offs, respectively, reflecting the resource strain of food production. Additionally, climate literature underrepresents social goals such as Peace, Justice, and Strong Institutions (SDG16), Gender Equality (SDG5), and Quality Education (SDG4), with 40% of their interactions classified as trade-offs. This suggests a bias toward environmental and economic dimensions, particularly from the Global North perspective, potentially marginalizing SDGs tied to equity and human vulnerability.

National planning must integrate PB-SDG feedback to avoid damaging trade-offs, such as observed land system (PB6) conflicts with food and water goals (SDG2/SDG6), by predicting resource pressures. We should prioritize underrepresented social goals, often negatively framed in climate studies, for equity and to build resilience for environmental action. We need to evaluate techno-optimism, such as biofuels potentially worsening land pressure (PB6) despite climate aims (SDG13), against firm biophysical limits. To reduce these trade-offs, we propose three steps: **(1)** Use integrated metrics, such as dashboards showing resource footprints alongside poverty/inequality, to reveal social-ecological links. **(2)** Strengthening governance, for instance, with scientific councils setting PB-based standards or budgets that guide SDG actions. **(3)** Ensure equity, for example, by using environmental tax revenues for Just Transition programs that support communities affected by respecting planetary boundaries.

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Exploring the interplay between Planetary Boundaries and Sustainable Development Goals using Large Language Models

Rhomrassi et al.

April 13, 2025

We retrieved 40,037 open-access articles from the OpenAlex database [5], based on their relevance to at least one of the nine PBs and analysed them using LLMs.

The keywords used where

- Climate change
- Ocean acidification
- Stratospheric Ozone Depletion
- Biogeochemical flow
- Global Freshwater Use
- Land System Change
- Biosphere integrity
- Novel entities
- Atmospheric Aerosol Loading

These articles were primarily obtained in PDF format, often as author-copy versions. To extract structured textual data, we utilized the AI-based library Grobid [1], which converted the documents into XML/TEI format. This process systematically removed irrelevant elements to the present study, such as figures, acknowledgments, and bibliographic sections, ensuring a clean and well-structured dataset for subsequent analysis.

For this study, we used Google's Gemini-1.5 Flash, following the approach validated in Larosa et at [3]. A key advantage of this model is its extensive one-million token context window, which enables the analysis of lengthy scientific articles in a single prompt. To mitigate potential biases from simple keyword matching, the model's knowledge was supplemented with explicit definitions of PBs and SDGs. The SDGs definition is extracted from Wikipedia ([4]). The PB definitions are summarized from the PB article [6].

The methodology involves sequentially applying five distinct queries to the LLM, with prompts detailed in subsequent sections. The processing pipeline for each individual article follows these steps:

- **SDG Allocation**: Assigns Sustainable Development Goals (SDGs) to the article without predefined constraints.
- **PB Allocation**: Identifies the relevant Planetary Boundaries for the article, also without constraints.
- **SDG-PB Relationship Classification**: Determines whether the interaction between each SDG-PB pair represents a synergy, a trade-off, or a neutral link.
- **SDG-PB Causality Analysis**: Establishes the directionality of the relationship, i.e., whether an SDG influences a PB or vice versa.

Despite the model's large context window, we imposed a limit of 20 SDG-PB pairs per query in Steps 3 and 4. Although a higher number is technically feasible, performance degradation occurs due to output token limitations, which adversely affects the quality of reasoning and results in less robust justifications. To ensure reliable identification of synergies and trade-offs, we required clear textual evidence of measurable actions, policies, or outcomes that directly demonstrate a relationship between an SDG and a PB. Additionally, the prompt explicitly discouraged speculative or overly broad assumptions, prioritizing evidence-based reasoning to strengthen the validity of the findings.

Finally, to enhance the classification of synergies and trade-offs, an additional fifth prompt was introduced to validate whether the identified interactions were genuinely reflected in the text, using an experimental reasoning model (Gemini 2.0 Flash Thinking, [2]). Here, synergy classifications were further split into three groups: "Generality", "Misled by Positivity", and "Actual Synergy". This step was essential for distinguishing actual synergies—where progress in one goal directly benefits another—from generalized statements and overly optimistic predictions shaped by the generally positive tone of scientific literature. Similarly, trade-offs were classified in three groups too: "Actual Trade-off", "Generic Negative Association", and "Double Negative (Co-Degradation)", to differentiate between actual trade-offs (where progress in one domain pressures another), cases of generic negative associations without providing evidence from the text, and co-degradations (where harm to one domain negatively affects another). We used this clarification to understand large groups of trade-offs or synergies, and to have a better interpretation of the results.

Prompts

Prompt 1

In this prompt we identify the SDGs. For every SDG, the definition is as the one for SDG1 that we copy below

SDG1: SDG1) NO POVERTY: Sustainable Development Goal 1 (SDG 1 or Global Goal 1), one of the 17 Sustainable Development Goals established by the United Nations in 2015,

calls for the end of poverty in all forms. The official wording is: "No Poverty". Member countries have pledged to "Leave No One Behind": underlying the goal is a "powerful commitment to leave no one behind and to reach those farthest behind first". SDG 1 aims to eradicate every form of extreme poverty including the lack of food, clean drinking water, and sanitation. Achieving this goal includes finding solutions to new threats caused by climate change and conflict. SDG 1 focuses not just on people living in poverty, but also on the services people rely on and social policy that either promotes or prevents poverty. The goal has seven targets and 13 indicators to measure progress. The five outcome targets are: eradication of extreme poverty; reduction of all poverty by half; implementation of social protection systems; ensuring equal rights to ownership, basic services, technology and economic resources; and the building of resilience to environmental, economic and social disasters. The two targets related to means of implementation SDG 1 are mobilization of resources to end poverty; and the establishment of poverty eradication policy frameworks at all levels. Despite the ongoing progress, 10 percent of the world's population live in poverty and struggle to meet basic needs such as health, education, and access to water and sanitation. Extreme poverty remains prevalent in low-income countries, particularly those affected by conflict and political upheaval. In 2015, more than half of the world's 736 million people living in extreme poverty lived in Sub-Saharan Africa. The rural poverty rate stands at 17.2 percent and 5.3 percent in urban areas (in 2016). One of the key indicators that measure poverty is the proportion of population living below the international and national poverty line. Measuring the proportion of the population covered by social protection systems and living in households with access to basic services is also an indication of the level of poverty.

```
definitionsseparated = [
str("""SDG1) NO POVERTY: ..."""),
str("""SDG2) ZERO HUNGER: ..."""),
str("""SDG3) GOOD HEALTH AND WELL-BEING: ..."""),
str("""SDG4) QUALITY EDUCATION: ..."""),
str("""SDG5) GENDER EQUALITY: ..."""),
str("""SDG6) CLEAN WATER AND SANITATION: ..."""),
str("""SDG7) AFFORDABLE AND CLEAN ENERGY: ..."""),
str("""SDG8) DECENT WORK AND ECONOMIC GROWTH: ..."""),
str("""SDG9) INDUSTRY, INNOVATION AND INFRASTRUCTURE: ..."""),
str("""SDG10) REDUCED INEQUALITIES:  ..."""),
str("""SDG11) SUSTAINABLE CITIES AND COMMUNITIES: ..."""),
str("""SDG12) RESPONSIBLE PRODUCTION AND CONSUMPTION: ..."""),
str("""SDG13) CLIMATE ACTION: ..."""),
str("""SDG14) LIFE BELOW WATER: ..."""),
str("""SDG15) LIFE ON LAND: ..."""),
str("""SDG16) PEACE, JUSTICE AND STRONG INSTITUTIONS: ..."""),
str("""SDG17) PARTNERSHIPS FOR THE GOALS: ...""")         ]
prompt = f"""
        You are tasked with identifying explicit links to different Sustainable
        Development Goals (SDGs) in a corpus of text. Your goal is to identify
```

the SDGs in each text strictly adhering to this list following list:

```
{definitionsseparated[0]}
{definitionsseparated[1]}
{definitionsseparated[2]}
{definitionsseparated[3]}
{definitionsseparated[4]}
{definitionsseparated[5]}
{definitionsseparated[6]}
{definitionsseparated[7]}
{definitionsseparated[8]}
{definitionsseparated[9]}
{definitionsseparated[10]}
{definitionsseparated[11]}
{definitionsseparated[12]}
{definitionsseparated[13]}
{definitionsseparated[14]}
{definitionsseparated[15]}
{definitionsseparated[16]}

Do not use any prior knowledge of the SDGs, but strictly adhere to
this list.

**Original text**

Text: "{text}"

**Output Format:** Please, do not add any additional lines or comments
and do not change the given output format below even by one single line.
Do not write the same SDG more than once.In your Reason, do not only
quote part of the text but discuss how it is explicitly linked to the
SDG you provide as a response. If a text has absolutely no link to any
of the SDGs, write "0".

SDG: relevant SDG
Reason: clear justification as to how the SDG is explicitly present
within the given text, with key sentences that show the link between
the text measures and the SDG description

SDG: relevant SDG

Reason: clear justification as to how the SDG is explicitly present
within the given text, with key sentences that show the link between
the text measures and the SDG description
```

```
          ... and so on with as many identified explicit SDGs within the text.
        """
```

Prompt 2

```
pbdef = f"""
PB: **PB1) Climate Change**
The alteration of the Earth's climate system due to human activities, primarily
through the emission of greenhouse gases like carbon dioxide ($\rm{CO_2}$) that
trap heat in the atmosphere.

**Control Variables:**
- **Atmospheric $\rm{CO_2}$ concentration** (target: below 350 ppm)
- **Change in radiative forcing** (target: less than +1 W/m² compared to
preindustrial levels)

PB: **PB2) Ocean Acidification**
The ongoing decrease in the pH of the Earth's oceans, caused by the uptake of
anthropogenic $\rm{CO_2}$ from the atmosphere. This process makes it harder for
marine organisms to build and maintain shells and skeletons made of calcium carbonate.

**Control Variable:**
- **Average global surface ocean saturation state with respect to aragonite
($\rm{\Omega}$arag)** (target: $\rm{\Omega}$arag maintained at or above 80\% of
the pre-industrial level)

PB: **PB3) Stratospheric Ozone Depletion**
The thinning of the ozone layer in the stratosphere due to human-produced
chemicals like chlorofluorocarbons (CFCs), which increases the amount of harmful
ultraviolet (UV) radiation reaching the Earth's surface.

**Control Variable:**
- **Stratospheric $\rm{O_3}$ concentration** (target: less than 5\% reduction from
the pre-industrial level of 290 Dobson Units)

PB: **PB4) Biogeochemical Flows (formerly Phosphorus and Nitrogen Cycles)**
The disruption of the natural cycles of essential elements like phosphorus (P) and
nitrogen (N) due to human activities, leading to eutrophication of water bodies
and other detrimental effects.

**Control Variables:**

- **Phosphorus:**
  - **Global:** P flow from freshwater systems to the ocean (target: less than 11
```

Tg P/year)
- **Regional:** P flow from fertilizers to erodible soils (target: less than 6.2 Tg P/year)

- **Nitrogen:**
  - **Global:** Industrial and intentional biological fixation of N (target: less than 62 Tg N/year)

PB: **PB5) Freshwater Use**
The consumption of freshwater resources by humans for various purposes, such as agriculture, industry, and domestic use, potentially impacting the availability of water for ecosystems and other users.

Control Variables:
- **Global:** Maximum amount of consumptive blue water use (target: less than 4000 $\rm{km^3}$/year)
- **Basin-specific:** Blue water withdrawal as a percentage of mean monthly river flow, with different thresholds for low, intermediate, and high flow periods

PB: **PB6) Land-System Change**
The conversion of natural land cover, such as forests and grasslands, to agricultural land, urban areas, and other human-modified landscapes, impacting the biogeophysical processes that regulate climate.

Control Variables:
- **Global:** Area of forested land as a percentage of original forest cover (target: above 75\%)
- **Biome-specific:** Area of forested land as a percentage of potential forest cover, varying for different forest types

PB: **PB7) Change in Biosphere Integrity (formerly Biodiversity Loss)**
The impact of human activities on the variety of life on Earth, encompassing genetic, species, and ecosystem diversity, influencing the biosphere's resilience and functioning.

Control Variables:
- **Genetic Diversity:** Extinction rate (target: less than 10 extinctions per million species-years)
- **Functional Diversity:** Biodiversity Intactness Index (BII) (target: BII maintained at 90\% or above)

PB: **PB8) Introduction of Novel Entities (formerly Chemical Pollution)**
The release of new substances, new forms of existing substances, and modified life forms into the environment, with potential for unknown and harmful effects on the Earth system.

```
**Control Variable:**
- No specific control variable defined due to the complex and diverse nature of
novel entities

PB: **PB9) Atmospheric Aerosol Loading**
The increase in the concentration of aerosol particles in the atmosphere due to
human activities, influencing air quality, cloud formation, and regional climate
patterns.

**Control Variable:**
- **Aerosol Optical Depth (AOD)** measured regionally (currently quantified for
the South Asian Monsoon region)
"""

prompt = f"""
        You are tasked with analyzing a given text to identify explicit links to
        planetary boundaries (PBs). Your goal is to find strong, well-justified
        connections between the text and the planetary boundaries, avoiding weak
        or overly generalized associations. Your response must focus strictly on
        the given text and provide clear reasoning supported by key sentences from
        the text itself.

        The list of PBs you must choose from is: [PB1) Climate Change, PB2) Ocean
        Acidification, PB3) Stratospheric Ozone Depletion, PB4) Biogeochemical
        Flows (formerly Phosphorus and Nitrogen Cycles), PB5) Freshwater Use, PB6)
        Land-System Change, PB7) Change in Biosphere Integrity (formerly
        Biodiversity Loss), PB8) Introduction of Novel Entities (formerly Chemical
        Pollution), PB9) Atmospheric Aerosol Loading]

        Below you have the strict definitions for each of the PBs:

{pbdef}

        Do not use any prior information or external knowledge on planetary
        boundaries. Your response must be based only on the provided text
        **below**, avoiding general statements. Please, do not confuse content
        from the PB definitions (above) and the text that you have to analyze
        (below).

        **Original text**

        Text: "{text}"

        **Output Format:** Please, do not add any additional lines or comments and
```

```
do not change the given output format below even by one single line. Do
not write the same planetary boundary more than once. In your Reason, do
not only quote part of the text but discuss how it is explicitly linked to
the planetary boundary you provide as a response. If a text has absolutely
no link to any planetary boundary, write ONLY "0".


  PB: Relevant PB
  Reason: Clear justification as to how the PB is explicitly present or
  addressed within the given text, with key sentences that show the link
  between the text's measures and the PB description

  PB: Relevant PB
  Reason: Clear justification as to how the PB is explicitly present or
  addressed within the given text, with key sentences that show the link
  between the text's measures and the PB description

  ... and so on with as many identified explicit PBs within the text.
"""
```

Prompt 3

Once we know every SDG and PB related to a paper, we add this information to the prompt and find the interaction between each SDG-PB pair represents a synergy, a trade-off, or a neutral link

```
prompt = f"""
**Instructions**
You are analyzing the relationship between Sustainable Development Goals (SDGs)
and Planetary Boundaries (PBs) based on the given text. Your task is to determine
whether the relationship between each SDG-PB pair is a synergy, a trade-off, or
neutral, strictly based on evidence found in the provided text.

    - Synergy: A synergy must be explicitly stated or strongly implied in the
    text. Clear measures, actions, policies, or outcomes must be present in the
    text that demonstrate how achieving the SDG directly benefits the PB or vice
    versa. If you identify a synergy, cite the specific sentence(s) from the text
    that support this conclusion, as well as providing a strong reasoning of your
    own. Do not assume synergies based on general SDG discourse or the overly
    positive tone of scientific papers-only extract what is clearly supported in
    the text.
    - Trade-off: A trade-off exists if the measures, policies, or outcomes for
    achieving the SDG increase pressure on the PB or vice versa. The text must
    provide direct or indirect evidence of such a conflict. Additionally, if the
```

```
    text does not make the trade-off explicit, but there is well-established
    scientific evidence in widely recognized literature (e.g., IPCC reports, peer-
    reviewed studies), reference that literature while clearly explaining how it
    applies to the specific case in the text. Do not introduce speculative or
    overly broad assumptions-provide evidence.
    - Neutral: If the relationship between the SDG and PB is unclear, weak, or
    loosely implied without strong supporting evidence, classify it as neutral. If
    one or both elements (SDG/PB) appear in the text but are unrelated or only
    vaguely mentioned, classify as neutral. Provide a brief explanation of why the
    connection is weak, unclear, or unsupported.

**Original text to analyze**
Text: "{text}"
**End of Original text to analyze**

**SDG allocation to text**
{structuredoutputsdg}

**PB allocation to text**
{structuredoutputpb}

**Output Format:** Please, do not add any additional lines or comments and do not
change the given output format below even by one single line.

Pair: {sdg} - {pb}
Link: Synergy/Trade-off/Neutral
Reason: Reason for your choice, indicating clearly how the measures, objectives or
outcomes discussed within the text would result in a synergy, trade-off, or a
neutral link between the pair.
...
"""
```

Prompt 4

In this step, we want to know the causal relationship between PB and SDGs.

```
prompt = f"""
**Instructions**
You are analyzing the causal relationship between Sustainable Development Goals
(SDGs) and Planetary Boundaries (PBs) based on the given text. Your task is to
determine which one of the SDG or PB is the cause and which one is the consequence
within the text, strictly based on evidence found in the provided text.

    - Cause: Identify which of the SDG or PB is the cause. The cause is the
    element that directly or indirectly initiates or drives the action or change
```

```
    that affects the other element.
    - Consequence: Identify which of the SDG or PB is the consequence. The
    consequence is the element that is directly or indirectly affected by the
    cause.
    - Justification: Provide a clear justification based on the text as to why one
    element is the cause and the other is the consequence. The justification
    should rely strictly on the content provided in the text and should not
    include general assumptions or references to external sources. Only
    conclusions that are clearly supported by the text should be presented.

The order in which the SDG and PB are presented in the pair is not relevant to the
classification; focus on the content of the text to determine which element acts
as the cause and which as the consequence.

**Original text to analyze**
Text: "{text}"
**End of Original text to analyze**

**SDG allocation to text**
{structuredoutputsdg}

**PB allocation to text**
{structuredoutputpb}

**Output Format:** Please, do not add any additional lines or comments and do not
change the given output format below even by one single line.
"""
```

Prompt 5:

This prompt has two questions. We ask for a confirmation of the previous steps, and distinguish synergies from double synergies and to identify trade-offs rom co-degradations.

```
\textit{Synergy analysis:}

prompt = f"""You are an expert analyst evaluating LLM responses about synergies
between Planetary Boundaries (PBs) and Sustainable Development Goals (SDGs),
specifically for the PB {pb} and SDG {sdg}.  You will be given a list of titles
and reasons for {linktype} links the LLM found in different scientific papers.
Your task is to analyze each title and reason and identify ONLY those that
represent 'actual synergies'.  An 'actual synergy' is where the reason
demonstrates a clear and direct link in the text showing how actions towards SDG
{sdg} demonstrably improve PB {pb} or vice versa, with a clear mechanism or causal
pathway described.  Reject reasons that are generalities or are misled by
```

```
positivity in the text (where the reason infers synergy without a strong textual
basis or due to positive framing alone).

For each title, classify it in 'Generality', 'Misled by Positivity', or 'Actual
Synergy'

**Criteria for 'Actual Synergy':** The 'Reason' must demonstrate a clear and direct
link in the text showing how promoting SDG {sdg} demonstrably improves PB {pb}, or
vice versa.  There should be a clear mechanism or causal pathway described in the
text.  Reject reasons that are generalities or are misled by positivity.
Generalities are broad, vague connections without specific mechanisms.
Misled-by-positivity cases infer synergy due to positive text framing but lack
explicit links or strong textual evidence.


**Output format:**
Title: ...
Class: 'Generality', 'Misled by Positivity', or 'Actual Synergy'
Reason: Explain your choice

Title: ...
Class: 'Generality', 'Misled by Positivity', or 'Actual Synergy'
Reason: Explain your choice

... and so on.

**Texts to analyze:**
{textcontent}
"""

\textit{Trade-off analysis:}

prompt = f"""You are an expert analyst evaluating LLM responses about trade-offs
between Planetary Boundaries (PBs) and Sustainable Development Goals (SDGs),
specifically for the PB {pb} and SDG {sdg}. You will be given a list of titles and
reasons for {linktype} links the LLM found in different scientific papers. Your
task is to analyze each title and reason and classify them into one of three
categories: 'Actual Trade-off,' 'Generic Negative Association,' or 'Double Negative.'

    **Key Criteria for Classification:**
    - **Actual Trade-off:** A true trade-off occurs when **PROGRESS in
    one domain directly creates pressure on the other**. The reason
    must describe a **clear mechanism** showing how actions benefiting
    SDG {sdg} actively **harm** PB {pb} (or vice versa).
```

- **Example of Actual Trade-off:** Expanding irrigation systems (improving SDG 2: Zero Hunger) increases freshwater consumption, worsening PB 5 (Global Freshwater Use).
- **Not a Trade-off:** If both PB {pb} and SDG {sdg} are harmed but **not due to progress in one domain**, it is NOT a trade-off.

- **Generic Negative Association:** A broadly negative link without a **specific cause-effect trade-off mechanism**. It may describe **environmental or societal deterioration** without showing a **direct tension** between SDG {sdg} and PB {pb}.

- **Example:** "Climate change is bad for food security" is too vague because it does not explain how improving one aspect **worsens** another.

- **Double Negative (Co-Degradation):** This happens when **both PB {pb} and SDG {sdg} are harmed at the same time, but one does not cause the other**. In this case, the relationship is **not a trade-off but mutual decline.**
- **Example of Double Negative:** Ocean acidification (PB 2) harming marine biodiversity (SDG 14) is NOT a trade-off because SDG 14 is not improving-both are simply declining together.

- **Misclassification to avoid:** "Ocean acidification worsens marine biodiversity, therefore it is a trade-off" → **Wrong! This is a Double Negative, not a trade-off.**

Important: Reject cases where the reasoning falsely infers a trade-off simply because both PB {pb} and SDG {sdg} are negatively affected.

Output format:
Title: ...
Class: 'Actual Trade-off', 'Generic Negative Association', or 'Double Negative'
Reason: Explain your choice

Title: ...
Class: 'Actual Trade-off', 'Generic Negative Association', or 'Double Negative'
Reason: Explain your choice

Texts to analyze:

```
{textcontent}
"""
```

References

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[3] Francesca Larosa, Sergio Hoyas, H. Alberto Conejero, Javier Garcia-Martinez, Francesco Fuso Nerini, and Ricardo Vinuesa. Large language models in climate and sustainability policy: limits and opportunities, 2025. https://arxiv.org/abs/2502.02191.

[4] Francesca Larosa, Lamyae Rhomrassi, Sergio Hoyas, J. Alberto Conejero, Francesco Fuso Nerini, Javier García-Martínez, and Ricardo Vinuesa. Leveraging artificial intelligence to unravel SDG interlinkages and progress. *SSRN*, jan 2025. https://ssrn.com/abstract=5175765.

[5] OpenAlex. Openalex: The open catalog of the global research system, 2024. Accessed: 2024-03-04.

[6] Johan Rockström, Louis Kotzé, Svetlana Milutinović, Frank Biermann, Victor Brovkin, Jonathan Donges, Jonas Ebbesson, Duncan French, Joyeeta Gupta, Rakhyun E. Kim, Timothy Lenton, Dominic Lenzi, Nebojsa Nakicenovic, Barbara Neumann, Fabian Schuppert, Ricarda Winkelmann, Klaus Bosselmann, Carl Folke, Wolfgang Lucht, and Will Steffen. The planetary commons: A new paradigm for safeguarding earth-regulating systems in the anthropocene. *Proceedings of the National Academy of Sciences*, 121:e2301531121, 01 2024.